

VLANs & Statik Routing

1. In Cisco Packet Tracer, create a simple two-switch network with three VLANs (e.g., VLAN 10 for 'Foodies', VLAN 20 for 'Gamers', VLAN 30 for 'Shoppers'), assign three PCs to each VLAN, and verify that PCs in the same VLAN can ping each other but not PCs in other VLANs.

->Here is a **Cisco Packet Tracer** setup you can follow.

1. Network Topology

Use:

- **2 × Cisco 2960 switches**
- **9 × PCs**
- 3 PCs for each VLAN

Trunk Link

SW1 ===== SW2

/ | \ / | \

/ | \ / | \

PC1 PC2 PC3 PC4 PC5 PC6

VLAN10 VLAN10

PC7 PC8 PC9 (connect according to VLAN assignments)

A cleaner assignment is:

VLAN Name	PCs
-----------	-----

10	Foodies PC1, PC2, PC4
----	-------------------------

20	Gamers PC3, PC5, PC7
----	------------------------

30	Shoppers PC6, PC8, PC9
----	------------------------

2. Create VLANs on Both Switches

On **SW1**:

enable

configure terminal

vlan 10

name Foodies

exit

vlan 20

name Gamers

exit

vlan 30

name Shoppers

exit

Do the same on **SW2**.

3. Assign Access Ports

For example, on SW1:

interface range fa0/1-2

switchport mode access

switchport access vlan 10

exit

interface fa0/3

switchport mode access

switchport access vlan 20

exit

On SW2, assign the remaining PCs according to their VLAN.

For example:

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

```
interface fa0/2
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

```
interface fa0/3
```

```
switchport mode access
```

```
switchport access vlan 30
```

```
exit
```

Repeat the assignments until all 9 PCs are connected to their appropriate VLANs.

4. Configure the Switch-to-Switch Link as a Trunk

Suppose SW1 and SW2 are connected using Fa0/24.

On **both switches**:

```
interface fa0/24
```

```
switchport mode trunk
```

```
exit
```

The trunk allows VLAN 10, 20, and 30 traffic to travel between the switches.

5. Assign IP Addresses to PCs

Use different IP addresses for each VLAN:

VLAN 10 – Foodies

PC	IP Address	Mask
----	------------	------

PC1	192.168.10.11	255.255.255.0
-----	---------------	---------------

PC2	192.168.10.12	255.255.255.0
-----	---------------	---------------

PC4	192.168.10.13	255.255.255.0
-----	---------------	---------------

VLAN 20 – Gamers

PC	IP Address	Mask
----	------------	------

PC3	192.168.20.11	255.255.255.0
-----	---------------	---------------

PC5	192.168.20.12	255.255.255.0
-----	---------------	---------------

PC7	192.168.20.13	255.255.255.0
-----	---------------	---------------

VLAN 30 – Shoppers

PC	IP Address	Mask
----	------------	------

PC6	192.168.30.11	255.255.255.0
-----	---------------	---------------

PC8	192.168.30.12	255.255.255.0
-----	---------------	---------------

PC9	192.168.30.13	255.255.255.0
-----	---------------	---------------

No default gateway is required because the goal is to demonstrate **same-VLAN communication without inter-VLAN routing**.

6. Verify the VLANs

On each switch:

show vlan brief

You should see:

10 Foodies

20 Gamers

30 Shoppers

Check the trunk:

show interfaces trunk

7. Test Connectivity

From **PC1 (VLAN 10)**:

ping 192.168.10.12

✓ Should succeed because both PCs are in VLAN 10.

Try:

ping 192.168.20.11

✗ Should fail because PC3 is in VLAN 20.

Try:

ping 192.168.30.11

✗ Should fail because PC6 is in VLAN 30.

Expected Results

Test	Expected
PC1 → PC2	✓ Success
PC1 → PC4	✓ Success
PC3 → PC5	✓ Success
PC3 → PC7	✓ Success
PC6 → PC8	✓ Success
PC6 → PC9	✓ Success
VLAN 10 → VLAN 20	✗ Fail
VLAN 10 → VLAN 30	✗ Fail
VLAN 20 → VLAN 30	✗ Fail

Conclusion: The two-switch network successfully separates the three groups using VLANs. PCs belonging to the **same VLAN** can communicate across the trunk, while PCs in **different VLANs cannot communicate** because no router or Layer-3 switch has been configured for inter-VLAN routing.

2. Configure trunk ports between the two switches in your Packet Tracer network so that all VLANs can communicate across both switches.

Hint: Use the 'switchport mode trunk' command and allow all VLANs on the trunk link.

->Configure Trunk Ports Between Two Switches

Assume the two switches are connected through **FastEthernet 0/24**.

1. Configure the trunk on SW1

Open **SW1** → **CLI** and enter:

enable

configure terminal

interface fastethernet 0/24

switchport mode trunk

switchport trunk allowed vlan all

no shutdown

exit

end

write memory

2. Configure the trunk on SW2

Open **SW2** → **CLI** and enter the same configuration:

enable

configure terminal

```
interface fastethernet 0/24
switchport mode trunk
switchport trunk allowed vlan all
no shutdown
exit
```

```
end
```

```
write memory
```

3. Verify the Trunk

On both switches, run:

```
show interfaces trunk
```

You should see something similar to:

Port	Mode	Status	Native vlan
Fa0/24	on	trunking	1

Vlans allowed on trunk

```
Fa0/24 1-4094
```

4. Verify the VLANs

Run:

```
show vlan brief
```

Both switches should contain:

```
VLAN 10 Foodies
```

```
VLAN 20 Gamers
```

```
VLAN 30 Shoppers
```

5. Test Communication

Because the trunk carries VLAN 10, 20, and 30 between the switches:

- **VLAN 10 PC on SW1 → VLAN 10 PC on SW2:** ✓ Ping should succeed.
- **VLAN 20 PC on SW1 → VLAN 20 PC on SW2:** ✓ Ping should succeed.
- **VLAN 30 PC on SW1 → VLAN 30 PC on SW2:** ✓ Ping should succeed.
- **VLAN 10 → VLAN 20:** ✗ Still fails without inter-VLAN routing.
- **VLAN 10 → VLAN 30:** ✗ Still fails without inter-VLAN routing.

Conclusion

The Fa0/24 links on both switches are configured as **trunk ports** and allow all VLANs. This enables **VLAN 10 (Foodies)**, **VLAN 20 (Gamers)**, and **VLAN 30 (Shoppers)** to extend across both switches while keeping the VLANs logically separated.

3. Add a router to your Packet Tracer network and configure Inter-VLAN routing so that a PC in VLAN 10 can successfully ping a PC in VLAN 20.

Hint: Use router-on-a-stick configuration with sub-interfaces for each VLAN.

->Inter-VLAN Routing Using Router-on-a-Stick

We will add a router and use **subinterfaces** so VLAN 10 and VLAN 20 can communicate.

1. Topology

Connect the router to **SW1**:

PCs — SW1 == SW2

|

|

R1

Assume:

- Router interface: GigabitEthernet0/0
- SW1 router-facing port: Fa0/23
- SW1 ↔ SW2 trunk: Fa0/24

2. Configure SW1 Router Port as a Trunk

On SW1:

enable

configure terminal

interface fa0/23

switchport mode trunk

switchport trunk allowed vlan 10,20,30

no shutdown

exit

end

write memory

3. Configure the Router

On R1:

enable

configure terminal

interface gigabitEthernet 0/0

no shutdown

exit

Create the VLAN 10 subinterface:

interface gigabitEthernet 0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

exit

Create the VLAN 20 subinterface:

```
interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

Create VLAN 30 as well:

```
interface gigabitEthernet 0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit
```

```
end
```

```
write memory
```

4. Configure PC Default Gateways

The PCs need to use the router's subinterface as their gateway.

VLAN 10 – Foodies

PC	IP	Subnet Mask	Gateway
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.10.12	255.255.255.0	192.168.10.1

VLAN 20 – Gamers

PC	IP	Subnet Mask	Gateway
PC3	192.168.20.11	255.255.255.0	192.168.20.1
PC5	192.168.20.12	255.255.255.0	192.168.20.1

5. Verify Router Configuration

On R1:

```
show ip interface brief
```

You should see:

```
GigabitEthernet0/0.10  192.168.10.1  up  up
```

```
GigabitEthernet0/0.20  192.168.20.1  up  up
```

```
GigabitEthernet0/0.30  192.168.30.1  up  up
```

6. Test Inter-VLAN Communication

From **PC1 (VLAN 10)**, open Command Prompt:

```
ping 192.168.20.11
```

Expected:

```
Reply from 192.168.20.11
```

```
Reply from 192.168.20.11
```

```
Reply from 192.168.20.11
```

```
Reply from 192.168.20.11
```

✅ **Ping succeeds.**

You can also test the gateway:

```
ping 192.168.10.1
```

and from PC3:

```
ping 192.168.20.1
```

Why It Works

The router receives VLAN-tagged traffic through the trunk and uses its subinterfaces to route between the networks:

VLAN 10

192.168.10.0/24

|

↓

192.168.10.1

R1

192.168.20.1

|

↓

VLAN 20

192.168.20.0/24

Conclusion: Router-on-a-stick successfully enables communication between VLAN 10 and VLAN 20 while keeping them as separate Layer-2 VLANs.

4. Set up static routes on the router so that a new network segment (e.g., a 'Creators' VLAN 40 on a third switch) can communicate with all existing VLANs, and test connectivity with ping.

->Static Routing for VLAN 40 – Creators

Assume the existing network is:

- VLAN 10 — Foodies: 192.168.10.0/24
- VLAN 20 — Gamers: 192.168.20.0/24
- VLAN 30 — Shoppers: 192.168.30.0/24
- VLAN 40 — Creators: 192.168.40.0/24

Because VLAN 40 is connected to a **third switch**, we can use a separate router interface/link and a static route.

1. Create VLAN 40 on Switch 3

On **SW3**:

enable

configure terminal

vlan 40

name Creators

exit

Assign the PC port to VLAN 40:

```
interface fa0/1
switchport mode access
switchport access vlan 40
no shutdown
exit
```

2. Configure the Link Between SW3 and the Router

For this example, use:

R1 G0/1 ----- SW3 Fa0/24

Configure SW3:

```
interface fa0/24
switchport mode access
switchport access vlan 40
no shutdown
exit
```

Configure the router:

```
enable
configure terminal
```

```
interface gigabitEthernet 0/1
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
```

If VLAN 40 is directly connected to R1 through this interface, a static route for VLAN 40 is **not actually required**; R1 automatically knows 192.168.40.0/24 as a connected network. Static routes are needed when the new network is reached through **another router**.

3. Example With a Separate Router

To demonstrate an actual static route, suppose SW3 is connected to **R2**, and R2 connects to R1:

VLAN 40

|

SW3

|

R2

|

| 10.0.0.0/30

|

R1

/|\

V10 V20 V30

Configure R2:

```
interface gigabitEthernet 0/0
```

```
ip address 192.168.40.1 255.255.255.0
```

```
no shutdown
```

```
interface gigabitEthernet 0/1
```

```
ip address 10.0.0.2 255.255.255.252
```

```
no shutdown
```

On **R1**, add a static route to VLAN 40:

```
ip route 192.168.40.0 255.255.255.0 10.0.0.2
```

On **R2**, add routes back to the existing VLANs:

```
ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

```
ip route 192.168.20.0 255.255.255.0 10.0.0.1
```

```
ip route 192.168.30.0 255.255.255.0 10.0.0.1
```

4. Configure the Creators PC

For example:

Setting	Value
IP Address	192.168.40.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.40.1

5. Verify the Routes

On R1:

```
show ip route
```

You should see a static route similar to:

```
S 192.168.40.0/24 [1/0] via 10.0.0.2
```

On R2:

```
show ip route
```

6. Test With Ping

From the **Creators PC**:

```
ping 192.168.10.11
```

```
ping 192.168.20.11
```

```
ping 192.168.30.11
```

Expected:

```
Reply from 192.168.10.11
```

```
Reply from 192.168.20.11
```

```
Reply from 192.168.30.11
```

Then test from an existing VLAN back to Creators:

```
ping 192.168.40.10
```

 The ping should succeed.

Final Result

Source	Destination	Expected
Creators VLAN 40	Foodies VLAN 10	✓ Success
Creators VLAN 40	Gamers VLAN 20	✓ Success
Creators VLAN 40	Shoppers VLAN 30	✓ Success
Foodies VLAN 10	Creators VLAN 40	✓ Success
Gamers VLAN 20	Creators VLAN 40	✓ Success
Shoppers VLAN 30	Creators VLAN 40	✓ Success

Conclusion: Static routes provide a path between the new **Creators VLAN 40** network and the existing VLAN networks. Ping tests confirm end-to-end connectivity.

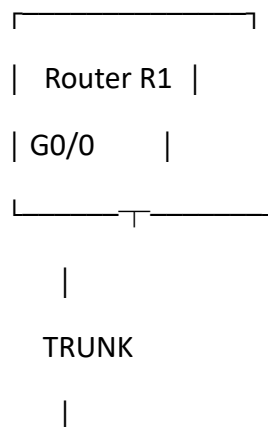
5. Design a mini network in Packet Tracer that mimics a Zomato-style office: create VLANs for 'Support', 'Delivery', and 'Management', configure trunking and static routing, and document your topology with a screenshot and a brief explanation of how devices in different VLANs communicate.

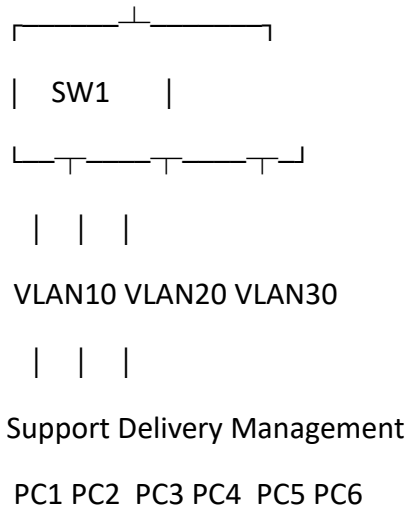
->Zomato-Style Office Network in Cisco Packet Tracer

Below is a simple design using **3 VLANs, trunking, a router-on-a-stick setup, and static routing.**

1. Network Topology

Create this topology in Packet Tracer:





2. VLAN Plan

VLAN	Department	Network	Gateway
10	Support	192.168.10.0/24	192.168.10.1
20	Delivery	192.168.20.0/24	192.168.20.1
30	Management	192.168.30.0/24	192.168.30.1

Use two PCs per department.

3. Configure VLANs on SW1

enable

configure terminal

vlan 10

name Support

exit

vlan 20

name Delivery

exit

vlan 30

name Management

exit

Assign PC ports:

interface range fa0/1-2

switchport mode access

switchport access vlan 10

exit

interface range fa0/3-4

switchport mode access

switchport access vlan 20

exit

interface range fa0/5-6

switchport mode access

switchport access vlan 30

exit

4. Configure the Trunk

Assume Fa0/24 connects SW1 to R1.

interface fa0/24

switchport mode trunk

switchport trunk allowed vlan 10,20,30

no shutdown

exit

5. Configure Router R1

Connect R1's G0/0 to SW1's Fa0/24.

enable

configure terminal

```
interface gigabitEthernet 0/0
```

```
no shutdown
```

```
exit
```

Create the VLAN subinterfaces:

```
interface gigabitEthernet 0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
exit
```

```
interface gigabitEthernet 0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
exit
```

```
interface gigabitEthernet 0/0.30
```

```
encapsulation dot1Q 30
```

```
ip address 192.168.30.1 255.255.255.0
```

```
exit
```

6. Configure Static Routing

Because all three VLAN networks are **directly connected to R1**, R1 automatically learns them as connected routes. Therefore, **no additional static routes are required between these three VLANs.**

If your assignment specifically requires a static route, add another router/network. For example, if an external network 192.168.40.0/24 is behind R2:

```
ip route 192.168.40.0 255.255.255.0 10.0.0.2
```

This demonstrates an actual static route while R1 continues to route between the three office VLANs.

7. Configure PC Addresses

Support

PC	IP Address	Gateway
----	------------	---------

PC1	192.168.10.10	192.168.10.1
-----	---------------	--------------

PC2	192.168.10.11	192.168.10.1
-----	---------------	--------------

Delivery

PC	IP Address	Gateway
----	------------	---------

PC3	192.168.20.10	192.168.20.1
-----	---------------	--------------

PC4	192.168.20.11	192.168.20.1
-----	---------------	--------------

Management

PC	IP Address	Gateway
----	------------	---------

PC5	192.168.30.10	192.168.30.1
-----	---------------	--------------

PC6	192.168.30.11	192.168.30.1
-----	---------------	--------------

Subnet mask for all PCs:

255.255.255.0

8. Test Inter-VLAN Communication

From PC1:

```
ping 192.168.20.10
```

From PC1 to Management:

```
ping 192.168.30.10
```

From PC3 to Support:

ping 192.168.10.10

Expected result:

Reply from 192.168.x.x

✅ Communication should succeed.

How Communication Works

Support PC

192.168.10.10

|



SW1

|

802.1Q Trunk

|



R1

|

├— VLAN 10 → 192.168.10.0/24

├— VLAN 20 → 192.168.20.0/24

└— VLAN 30 → 192.168.30.0/24

|



SW2

|



Delivery PC

192.168.20.10

The **switch** handles VLAN separation at Layer 2. When a device needs to communicate with another VLAN, traffic is sent to its **default gateway on R1**. R1 routes the packet between the VLAN networks and sends it back through the trunk to the appropriate VLAN.

Screenshot for Your Assignment

After completing the topology in Packet Tracer:

1. Arrange the router, switch, and six PCs neatly.
2. Add labels: **Support VLAN 10, Delivery VLAN 20, Management VLAN 30**.
3. Make sure the links are green.
4. Use **Capture/Forward** or successful ping results to demonstrate connectivity.
5. Take a screenshot of the complete Packet Tracer topology and the successful ping.

Brief explanation to submit:

The Zomato-style office network uses three VLANs: Support (VLAN 10), Delivery (VLAN 20), and Management (VLAN 30). The switch uses access ports for the PCs and an 802.1Q trunk to carry all VLANs to the router. Router-on-a-stick subinterfaces provide gateways for each VLAN and allow inter-VLAN communication. Devices in different VLANs communicate by sending traffic to their default gateway, where the router routes the packet to the destination VLAN.